

Interfacial Tensions of Polyelectrolyte Multiphase Coacervation

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Cite This: *Macromolecules* 2024, 57, 9698–9710



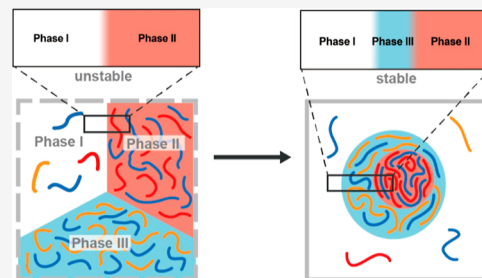
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ABSTRACT: Interfacial tension is a crucial property for a charged polymer system undergoing liquid–liquid phase separation as it governs the structure and morphology of condensate phases. While the interface problem of simple two-phase coacervation has been well documented, little light has been shed so far on the multiphase separation process. In this study, we theoretically investigate the interfacial properties in multiphase coacervation driven by the asymmetry of the linear charge density for different polycations. We calculate the density profiles of all species at these interfaces as well as the variations of interfacial tensions with salt concentration. The results indicate that in a three-phase coacervation system, direct contact of the dilute phase with the most condensed phase is unstable, so a middle layer consisting of a less condensed phase must appear. Then, a core–shell structure is formed, which is independent of the presence or absence of salt ions. Our study is in good agreement with the previous literature and provides deeper insight into understanding the spatial structure of multiphase coacervates.



INTRODUCTION

The formation of liquid biomolecular condensates depends on intracellular liquid–liquid phase separation (LLPS). The liquid condensates can be found in cells acting as a variety of organizations, such as membraneless organelles in cells, and they are fundamental in compartmentalizing the intracellular space.^{1–5} The LLPS usually relies on the complexation of the intrinsically disordered proteins, which cannot form well-defined three-dimensional structures without specific ligands or binding partners.⁶ As these IDPs are rich in polar and charged residues, the electrostatic interaction is believed to be one of the key driving forces for their complexation.⁷ The complex coacervation of oppositely charged polymers is in analogy with the intracellular condensation, which refers to the associative phase separation of polyelectrolytes, forming a dense polymer-rich phase together with a dilute phase.^{8–10}

Theoretically, the core of complex coacervation relies on the description of attractive electrostatic interactions, which leads to the condensation of charged species. The initial Voorn–Overbeek (VO) theory describes the electrostatic correlation based on Debye–Hückel correlation energy.⁹ Although it predicts the phenomenon of macrophase separation, this oversimplified theory fails to capture many important factors, especially the chain connectivity, and may lead to considerable deviation from the experimental results.^{11,12} The further developed random phase approximation (RPA) theory includes the chain connectivity when considering the electrostatic correlation.^{13–17} The RPA theory works well for weakly charged polyelectrolytes in a dense coacervate phase, where they adopt ideal Gaussian chain conformation.^{18–20} Nowadays, more theories have been developed to describe the bulk phase

behavior of coacervation to fit different experimental systems.^{21–30}

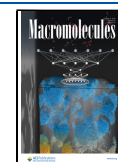
For liquid coacervate droplets, one of the most crucial properties is the interfacial tension between the coacervate phase and the dilute phase.³¹ Comparing to other LLPS processes, the electrostatic interaction-induced coacervate droplet shows an ultralow interfacial tension, mostly lower than 1 mN/m.^{32–36} The interfacial tension of membraneless organelles in vivo is even lower at $\sim 1 \mu\text{N/m}$.³⁷ This ultralow interfacial tension plays a key role in the deformability and dynamic properties of coacervates, highlighting the importance of its experimental and theoretical studies. The interfacial tension can be controlled by many factors, of which the salt concentration is the most widely discussed. Experiments have found that the interfacial tension γ will decrease with the increase of salt concentration c_s , and it follows the scaling law of $\gamma \propto (c_s^* - c_s)^{3/2}$, where c_s^* denotes the critical salt concentration,^{32,35} above which the phase separation will be suppressed. Other factors such as the temperature³⁵ or chain length³⁸ are also investigated. Much efforts have been made to give theoretical descriptions of the interfacial tension for the coacervate system.^{38–42} Based on the VO theory of the bulk coacervate phase, Qin et al. introduced the Cahn–Hilliard theory⁴³ to describe the interfacial free energy and derived the

Received: August 23, 2024

Revised: September 30, 2024

Accepted: October 7, 2024

Published: October 11, 2024



3/2 scaling law of tension dependence on salt concentration corresponding to the experimental results.⁴⁰ Lytle et al. further included the chain connectivity and the reversible adsorption process by use of the transfer matrix theory and observed the same scaling law.⁴¹ For an asymmetric polyelectrolyte system, Zhang and Wang studied the interfacial property based on the liquid-state theory and found that an electric double layer forms at the interface.⁴²

All of the above theoretical studies handle the simple two-phase coacervation process, namely, a dilute phase plus a dense phase. However, the intracellular LLPS always involves multiple components and usually leads to a multiphase separation, forming different coacervate phases coexisting with a dilute phase.^{37,44–47} The various liquid droplets provide distinct subcompartments for different biochemical reactions.⁴⁸ Besides, the structure and architecture of these multiphase droplets are of great importance since different subcompartments may play unique roles in the biological process, which relies on the core–shell architecture of the nucleolus.³⁷ The driving force for multiphase coacervation is still a rapidly growing topic. The separation between dense coacervate phases is usually attributed to the short-range interaction between different species parametrized by χ in theoretical treatment.^{49–51} On the other hand, the recent work highlights the electrostatic force-driven multiphase coacervation.^{52,53} If two kinds of polycations have distinct critical salt concentrations when they are complexed with polyanions separately, the electrostatic interaction will bring an effective repulsion between them and lead to the separation of two coacervate phases.⁵³ This process is illustrated in our previous theoretical work, where we have proved that the asymmetry of linear charge density or sequence of charged monomers may lead to the formation of multiphase coacervation.⁵⁴

For the multiphase separation, the interfacial properties between each two phases will be a more important issue than a simple two-phase separation process. When the solution are phase separated into different dense phases, the architecture and morphology of the small droplets depend on the tension of each interface.^{37,53,55,56} When one coacervate phase shows a relatively higher interfacial tension with the dilute phase, it is usually wrapped in the other coacervate phase and a core–shell structure forms. On the other hand, if the two coacervate phases have similar interfacial tension with the dilute phase whereas the interfacial tension between the two coacervate phases is lower, it usually tends to form a “Janus particle”-like droplet.⁵⁵ Only when the interfacial tension of the multiphase separation system is thoroughly studied, one can rationally design different structures of multiphase droplets for various applications.⁵⁶ For the phase separation process driven by short-range interaction, the interfacial tension has been well documented theoretically, and it usually follows the scaling law of $\gamma \propto (\chi - \chi_c)^{3/2}$, where χ_c denotes the Flory–Huggins parameter at the critical point.⁵⁷ However, for the multiphase complex coacervation process, the theory describing the interfacial tension between each two phases still lacks. In view of the fact that the multiphase separation can be driven by different types of interactions, the interface may have distinct properties and lead to different morphologies of coacervate droplets.

In this article, we investigate the interfacial properties when the polyelectrolyte solution undergoes multiphase coacervation driven by the asymmetry of linear charge density and give

some insights into the morphology of the coacervate droplets. We focus on the solution consisting of two types of polycations and one polyanion, where the two polycations are totally the same except for the different linear charge density. In our previous work, we have proved that the solution may undergo a 3-phase separation. Here, we further calculate their interfacial properties. First, we establish the theoretical framework for calculating the density profiles of each species and interfacial tension based on the RPA theory. Then, we study how the interfacial properties vary with various charge density asymmetries and with salt concentrations. For the interface between the dilute phase and the coacervate phase with higher charge density, we find that the density distribution for direct contact between the dilute phase and the more dense phase is not stable, and a middle layer composed of the less dense phase must appear between these two phases as a transition. We further discuss the unusual density profiles. Based on this result, we propose that the multiphase coacervation driven by charge density asymmetry will lead to a core–shell structure.

MODEL AND THEORY

Model. We consider a multicomponent polyelectrolyte system composed of two types of weakly charged polycations (denoted as A and B chains) and one polyanion C, the same as that used in our previous paper.⁵⁴ All polymers are Gaussian chains with the same Kuhn length b and the same degree of polymerization $N = 100$. Also, each type of monomer has the same volume of b^3 . We set $b = 1$ nm as the length unit. The charges are assumed to be distributed uniformly along each backbone (smeared model), and the linear charge density is denoted by σ_i ($i = A, B, C$). We set the charge densities of the A and C chains as $\sigma_A = 0.5$ and $\sigma_C = 0.5$, respectively. The small ions are point-like charges, and water molecules have a volume of b^3 as monomers. ϕ_i is the volume fraction of component i with $i = A, B, C, s$, and w , where letters represent polymer monomers A, B, and C, salt ions, and water molecules, respectively. The concentration ϕ_i is related to the particle density through $\phi_i(\mathbf{r}) = \rho_i(\mathbf{r})b^3$. We assume that all positive charges from polycations are compensated for by negative charges from polyanions at any position. Then, in salt-free systems, the local charge neutrality condition requires the relationship of $\phi_C(\mathbf{r})\sigma_C + \phi_A(\mathbf{r})\sigma_A - \phi_B(\mathbf{r})\sigma_B = 0$ since counterions are absent. In salt-containing systems, extra salt ions (each salt ion carries charge of $\pm e$) are added into the solutions and $\phi_s^+ = \phi_s^-$ is additionally required to guarantee electroneutrality. The counterions and salt ions are assumed to be likewise indistinguishable. For simplicity, all Flory–Huggins parameters are set to 0 to highlight the electrostatic interactions.

Our previous study shows that the system mentioned above may undergo macrophase separation.⁵⁴ A dilute phase coexists with two dense phases, and three interfaces appear simultaneously. The size of the droplet is much larger than the characteristic length of polymers, which makes it reasonable to ignore the curvature effect of interfaces. Assuming that z is the direction perpendicular to the interface and ignoring the density variation in the directions of x and y , it reduces to a one-dimensional interfacial problem.

Calculation of Density Profiles and Interfacial Tension. We first give the free energy expression at the interface in order to determine the interfacial tension between two coexisting phases. For a homogeneous bulk solution, its free energy density can be written as

$$f_b \equiv \frac{F_{\text{bulk}}}{Vk_B T} = f_{\text{entropy}} + f_{\text{short}} + f_{\text{elec}}, \quad (1)$$

where V is the volume of the system. f_{entropy} represents the translational entropy of all species, which is given by

$$f_{\text{entropy}} = \sum_i \frac{\phi_i}{N_i} \ln \phi_i \quad (2)$$

and f_{short} is the short-range non-electrostatic interaction among all species, that is

$$f_{\text{short}} = \frac{1}{2} \sum_{ij} \chi_{ij} \phi_i \phi_j \quad (3)$$

f_{elec} accounts for electrostatic correlation energy under the framework of RPA theory and it takes the following form

$$f_{\text{elec}} = \frac{b^3}{2} \int \frac{d\mathbf{k}}{(2\pi)^3} [\ln(1 + \sum_i \rho_i g_i^0(\mathbf{k}) u(\mathbf{k}) \sigma_i^2) - \sum_s \rho_s u(\mathbf{k}) \sigma_s^2] \quad (4)$$

Here, $g(\mathbf{k})$ represents the intramolecular monomer–monomer correlation function and its form is given in a previous publication.⁵⁴ $u(\mathbf{k}) = 4\pi l_B/k^2$ is the Coulomb interaction between two elementary charges. The Bjerrum length is $l_B = e^2/(\epsilon_0 \epsilon_r k_B T)$ with the vacuum permittivity of ϵ_0 and the relative solvent dielectric constant of ϵ_r . Considering the inhomogeneous solution with an interface, the local free energy can be extended, and it has the same expressions as eq 1 but with position-dependent concentrations. Especially, the electrostatic fluctuation contribution is approximated by 4 with all densities replaced by $\rho_i(\mathbf{r})$. This approximation has been employed by Fredrickson et al. when they developed an embedded fluctuation model to study the phase behavior of electrostatically complexed triblock copolymer gels.⁵⁸

Besides, the density heterogeneity at the interface causes a conformational entropy penalty and gives rise to an extra contribution to the free energy. When the interfacial width is not excessively narrow and the system is not far from the critical point, the interface is not very sharp and the densities change slowly over a length scale defined by the polymers. The term from the interaction potential between chain segments is much larger than the gradient term of the propagation function. Then, the weak inhomogeneity limit can be employed in this situation. The extra free energy contribution arising from the density gradient adopts a Landau form^{59,60}

$$f_{\text{inter}} = \sum_p \frac{b^2}{36\phi_p(\mathbf{r})} [\nabla \phi_p(\mathbf{r})]^2 \quad (5)$$

where p denotes polymer A, B, or C. For the one-dimension interface considered here, $\phi(\mathbf{r})$ reduces to $\phi(z)$.

The local electrostatic energy and the double electric layer effect, reported by Zhang et al.,⁴² are not considered here. Therefore, we assume that positive and negative charges are perfectly compensated for by each other at each location, leaving a net charge of zero everywhere. Combined with the aforementioned set of equations, the total free energy at the interface can be written as

$$\frac{F_{\text{inter}}}{k_B T} = \int_V (f_{\text{entropy}} + f_{\text{short}} + f_{\text{elec}} + f_{\text{inter}}) d\mathbf{r} \quad (6)$$

The free energy per unit area arising from the formation of the interface can be expressed by

$$\gamma \equiv \frac{\Delta F}{S} = \frac{1}{S} (F_{\text{inter}} - \sum_i \mu_i^b \bar{\phi}_i V) \quad (7)$$

where S is the interface area. $\bar{\phi}_i$ represents the average volume fraction of species i , and μ_i^b is the chemical potential in the bulk phase, defined as

$$\mu_i^b = f_b + \frac{\partial f_b}{\partial \phi_i} - \sum_j \phi_j \frac{\partial f_b}{\partial \phi_j} \quad (8)$$

The equilibrium interfacial properties are determined by minimizing the interfacial free energy with respect to the concentrations. Namely, the functional derivative of interfacial free energy with respect to each concentration should be equal to zero

$$\frac{\delta \gamma}{\delta \phi_i(\mathbf{r})} = 0 \quad (9)$$

Remembering that $\bar{\phi}_i V = \int_V \phi_i(\mathbf{r}) d\mathbf{r}$, we insert the expression of eq 7 into eq 9 and get the equations that the concentration distribution of each species should satisfy.

The equations for polymer concentrations are given by

$$\begin{aligned} \frac{1}{Sk_B T} \frac{\delta \gamma}{\delta \phi_p} &= \frac{1}{N_p} + \frac{1}{N_p} \ln \phi_p(z) + \sum_i \chi_{ip} \phi_i(z) \\ &+ \frac{1}{2} \int \frac{d\mathbf{q}}{(2\pi)^3} \frac{g_p(\mathbf{q}) u(\mathbf{q}) \sigma_p^2}{1 + \sum_i \rho_i(\mathbf{r}) g_i(\mathbf{q}) u(\mathbf{q}) \sigma_i^2} \\ &+ \frac{b^2}{36\phi_p^2(z)} \left(\frac{d\phi_p(z)}{dz} \right)^2 - \frac{b^2}{18\phi_p(z)} \cdot \frac{d^2 \phi_p(z)}{dz^2} \\ &- \mu_p^b = 0, \end{aligned} \quad (10)$$

where p denotes polymer A, B, or C. The salt ion distributions are given by

$$\begin{aligned} \frac{1}{Sk_B T} \frac{\delta \gamma}{\delta \phi_s} &= 1 + \ln \phi_s(z) + \sum_i \chi_{is} \phi_i(z) \\ &+ \frac{1}{2} \int \frac{d\mathbf{q}}{(2\pi)^3} \left[\frac{u(\mathbf{q}) \sigma_s^2}{1 + \sum_i \rho_i(\mathbf{r}) g_i(\mathbf{q}) u(\mathbf{q}) \sigma_i^2} \right. \\ &\left. - u(\mathbf{q}) \sigma_s^2 \right] - \mu_s^b = 0. \end{aligned} \quad (11)$$

For water, its concentration should satisfy

$$\frac{1}{Sk_B T} \frac{\delta \gamma}{\delta \phi_w} = 1 + \ln \phi_w + \sum_i \chi_{iw} \phi_i - \mu_w^b = 0 \quad (12)$$

With the constraints of charge neutrality and incompressibility, we obtain the relationship between different species. Then, ϕ_C and ϕ_w can be expressed as

$$\phi_C(z) = - \frac{\phi_A(z) \sigma_A + \phi_B(z) \sigma_B}{\sigma_C} \quad (13)$$

$$\phi_w(z) = 1 - \sum_p \phi_p(z) - \sum_s \phi_s(z) \quad (14)$$

Given that there are only three independent variables, eqs 10–12 can be further simplified. Finally, the equations are

$$\frac{\delta\gamma}{\delta\phi_{p'}} - \frac{\delta\gamma}{\delta\phi_c} \frac{\sigma_{p'}}{\sigma_c} - \left(1 - \frac{\sigma_{p'}}{\sigma_c}\right) \frac{\delta\gamma}{\delta\phi_w} = \frac{\delta\gamma}{\delta\phi_s} - \frac{\delta\gamma}{\delta\phi_w} = 0 \quad (15)$$

where $p' = A$ or B . Note that the above process is identical to the method of introducing the additional Lagrange multipliers for the constraint, eqs 13 and 14 in the interfacial free energy expression, eq 6.

The interfacial tension is the minimum interfacial free energy with the optimal density profiles. For simplicity, we still use γ to denote the interfacial tension in the following part. The procedures for calculating the interfacial tension are as follows. After multiphase separation, the compositions of each coexisting phase and the chemical potentials for all species are first obtained under the framework of RPA, which has been studied systematically in our previous work. All concentrations at the two coexisting phases are specified (bulk phases at two boundaries), which provide the boundary conditions of each interface, ensuring that the concentrations of all species at locations far from the interfaces are identical to those of the bulk phases. The density profiles at equilibrium are then solved by use of eq 15. Once the density profiles of all species are obtained, we can calculate the interfacial tensions by employing eq 7. The phase structure is further explored based on the interfacial tension.

RESULTS AND DISCUSSION

Interfacial Properties of the Two-Component Coacervation System. In order to validate the rationality of our theoretical framework in this article, the density profiles and interfacial properties of complex coacervation with simple two-component systems are first examined. We consider a polyelectrolyte solution consisting of polycations A, polyanions C, and salt ions $s^\pm$ with volume fractions $\phi_A(z) = \phi_C(z)$, $\phi_s^+ = \phi_s^- = \phi_s$ (polycations B are not involved in this case). The oppositely charged polymers A and C satisfy the charge neutrality condition so that their charges are compensated completely.

Figure 1 displays a typical phase diagram in the $\phi_A - \phi_s$ plane, which implies that the solution may undergo a macrophase separation into a polymer dense phase and a supernatant phase when it is in the region below the blue line (binodal line). The black dashed lines are tie lines connecting the compositions of the two coexisting phases. The polymer concentration in the coacervate phase, represented by the right branch of the binodal line, decreases with increasing salt concentration. Meanwhile, an increased polymer concentration is observed in the dilute phase, as indicated by the left branch of the blue line in Figure 1. Furthermore, on the $\phi_A - \phi_s$ plane, the two branches merge, indicating complete suppression of phase separation once the added salt concentration is beyond the critical salt concentration threshold at $\phi_s^* = 0.160$. The above results confirm the main features of typical two-component coacervation systems.

The density profiles of polycation A and salt ions at different added salt concentrations are displayed in Figure 2. As shown in Figure 2a, the concentration distribution of polyelectrolyte

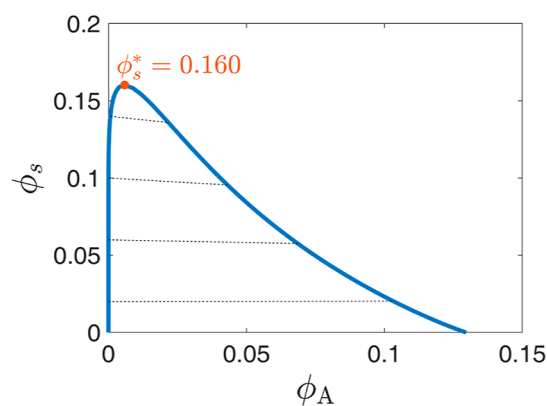


Figure 1. Phase diagram of a solution containing polycations A and polyanions C. $N_A = N_C = 100$, $\sigma_A = 0.50$, and $\sigma_C = -0.50$. The blue line is the binodal, and black dashed lines of the two-phase region are tie lines of phase separation, denoting the compositions of two coexisting phases. The red point corresponds to the critical salt concentration (CSC), $\phi_s^* = 0.160$, above which no phase separation occurs.

A follows a monotonically increasing trend from the supernatant phase (I-phase) to the coacervate phase (II-phase). With increasing salt concentration, the polymer concentration in the coacervate phase decreases, while the interfacial width increases, which is in accord with early studies.⁴¹ Addition of salt ions screens out the electrostatic interactions, and the weakened driving force of phase separation accounts for the broadened interface. However, a nonmonotonic distribution of salt concentration across the interface may be observed. At low content of added salts, $\phi_s^I = 0.02$ or 0.06 , salt ions arrive at a peak value at the interface. The local enrichment of salt ions vanishes at higher salt content, as shown by the purple line and the green line in Figure 2b. This nonmonotonic behavior in $\phi_s(z)$ has been reported in the previous work by Zhang et al.⁴² Furthermore, we also find that the salt-partitioning coefficient, which is the ratio of salt ions in the dense phase to the dilute phase, is larger than 1 at very low salt content and drops to a minimum below 1 at some intermediate concentration and increased further. This monotonic behavior comes from the chain connectivity correlation, as confirmed by Zhang et al.²⁴

The enrichment of salt ions at the interface is related to the conformational entropy loss of polymers. The interface properties are mainly determined by electrostatic interaction and chain conformation entropy. Electrostatic correlation drives the phase separation into two immiscible phases. Thus, stronger electrostatic correlation leads to a sharper interface. On the other hand, the chain conformation entropy loss is positively interrelated with the concentration gradient at the interface, so a flatter interface is favored. At low salt concentrations, strong electrostatic correlation between charged species induces a sharp interface. Therefore, the conformational entropy loss becomes very large. The enrichment of salt ions may enhance the electrostatic screening effect and allow for a broader interface, in favor of reducing the conformational entropy penalty. Upon adding more salt ions, Coulombic attractions become increasingly screened and the interfacial width becomes larger, in which case the conformational entropy penalty becomes less significant, while the translational entropy of salt ions dominates. Then, the enrichment effect at the interface disappears.

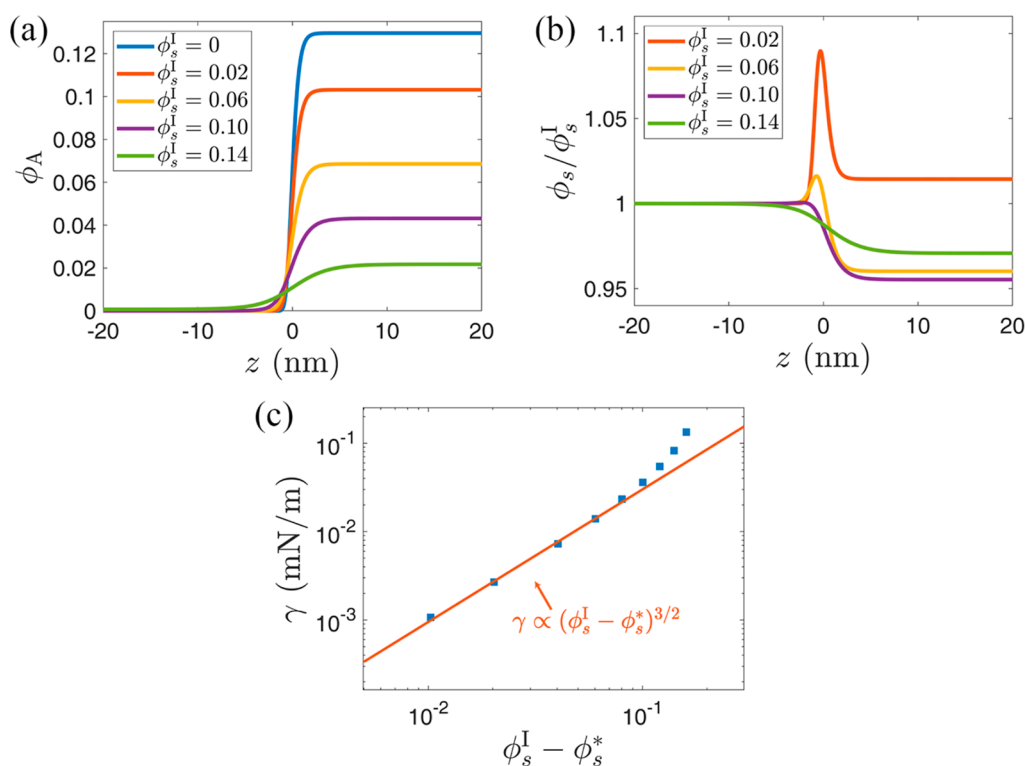


Figure 2. (a,b) Density profiles of different species for systems with increasing salt concentrations from the supernatant phase (I-phase) to the coacervate phase (II-phase). Density profiles of the polycations (a) and the salt ions (b). (c) Interfacial tensions γ as a function of the difference between supernatant salt concentration and the critical salt concentration $\phi_s^I - \phi_s^*$. Blue square symbols correspond to results obtained from theoretical prediction, with the red fit line showing the 1.5 scaling exponent near the critical point.

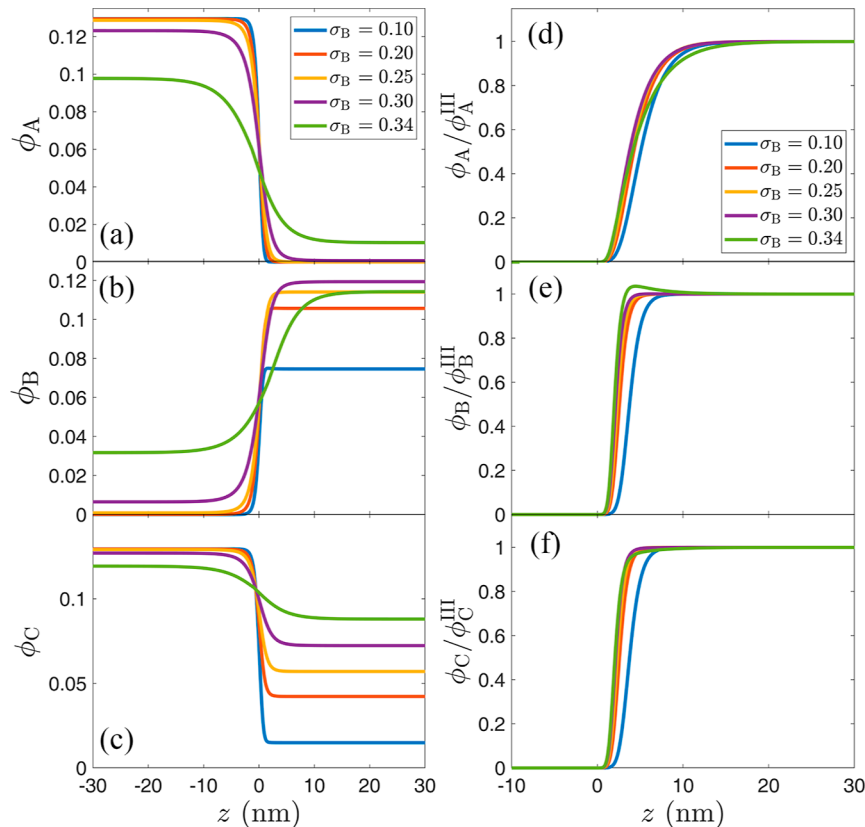


Figure 3. Distribution of different species with increasing σ_B . (a–c) Density profiles of different species from the AC-rich phase (II-phase) to BC-rich phase (III-phase). (d–f) Density profiles of different species from the supernatant phase (I-phase) to BC-rich phase (III-phase).

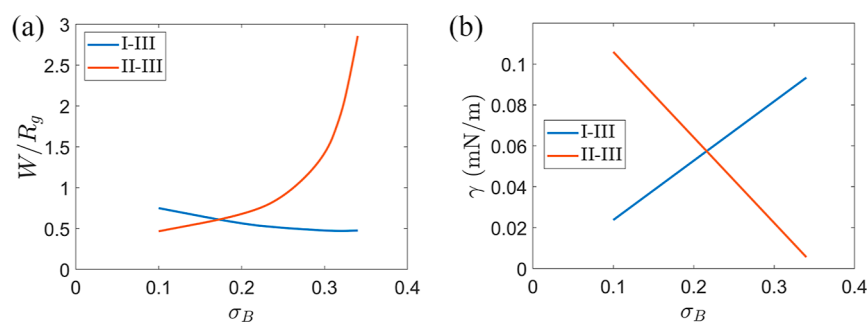


Figure 4. (a) Interfacial width between I and III phases (blue line) and II and III phases (magenta line) as a function of increasing σ_B . The profile of polyanion C is chosen to calculate the interfacial width and the vertical axis is plotted as the ratio of the interfacial width to the radius of gyration of the polyelectrolyte. (b) Interfacial tension as a function of σ_B .

We further investigated the interfacial tension between two phases. As shown in Figure 2c, the interfacial tension obtained from theoretical prediction is approximately on the order of $10^{-1} \sim 10^{-3}$ mN/m, which is in accord with the ultralow interfacial tension reported in experiments. We also find that the interfacial tension decreases with the increase in salt concentration in the supernatant phase ϕ_s^I . Being consistent with Qin's work, the red fitting line shows that γ follows a power law dependence of $\gamma \propto (\phi_s - \phi_s^*)^{3/2}$ near the critical point. The results based on our method are in good agreement with experiments and scaling behaviors of interfacial tension, which endorses the validity of our theoretical framework in predicting the interfacial properties of the coacervation system.

Interfacial Structure and Properties for the Multiphase Coacervation System. In this section, we concentrate on the interfaces between coexisting multiphases composed of multiple components. We first recall some main results obtained from a previous study of three-phase separation driven by electrostatic correlation, which is the basis for our subsequent discussion about the interfacial properties. For a salt-free multicomponent system containing only polycations A and B, polyanions C, and solvent w, multiphase separation can be driven solely by the differences in the charge density of A and B chains. When the linear charge density of B chain σ_B decreases and it is obviously smaller than σ_A , the originally stable solution will undergo three-phase separation. A dilute phase poor in polymers (phase I) coexists with a coacervate phase rich in A/C components (phase II) and another coacervate phase rich in B/C components (phase III). The critical point for the occurrence of three-phase separation is confirmed to be $\sigma_B = 0.354$. Addition of salt ions eventually results in the vanishing of the dilute phase, leaving the two coexisting AC-rich phase and BC-rich phase. However, salt appears to have little effect on regulating the effective immiscibility between the A and B chains. In the following, we handle the three different interfaces between the three phases.

Effects of Linear Charge Density Asymmetry on Interfacial Properties. For simplicity, we start from the salt-free system and vary σ_B from 0.10 to 0.34 while keeping the charge densities of other species as default values mentioned in the above section. Three coexisting phases as well as three interfaces appear at these values.

Figure 3 shows the density profiles of all polyelectrolytes at the interface between II and III phases at different σ_B values, which exhibit a monotonic variation trend. In addition, the distribution curves become smoother with increasing σ_B . We

attribute the slower change of the density profiles to the decreased effective repulsion effect between polycations A and B. Increase of σ_B leads to a decrease in charge density asymmetry as well as the driving force for multiphase separation and further increases the miscibility between A and B chains.

However, for the interface between the I-phase and III-phase, it displays some distinct properties. Figure 3e plots a sharper distribution of B monomer density than that of polymer A density (shown in Figure 3d). Especially, at $\sigma_B = 0.34$, the distribution of B monomers displays a non-monotonic trend with a peak value near the interface. The polycations with lower charge density tend to be concentrated near the interface, a phenomenon that may have the same underlying physical mechanisms as the enrichment of salt ions at the interface mentioned in the last subsection. The interfacial width is determined by the competition between the chain conformation entropy and electrostatic correlation strength. As electrostatic interaction becomes stronger, the interface tends to be sharper in order to reduce the extra free energy induced by electrostatic correlation. As shown in Figure 3e,f, higher charge density of B polymers has enhanced the electrostatic correlation effect and resulted in sharper density profile curves. However, a sharper interface incurs severe chain conformational entropy loss. Therefore, the weaker charge polymer B prefers residing near the interface to attenuate the electrostatic correlation effect, and the broader interface helps minimize the entropy penalty. This is similar to the role of salt ions in the case of a two-phase coacervation system.

Based on the obtained density profiles for each species, we calculate the interfacial width and the interfacial tension between two phases with varying charge density of polycation B. Two interfaces have different variations of interfacial width with σ_B (Figure 4a). With an increase in σ_B , the interfacial width between I and III phases becomes sharper gradually because of the enhanced electrostatic correlation effect. As for the II–III phase interface, the interfacial width monotonically increases with σ_B . When σ_B approaches 0.34, the interfacial width increases dramatically. In this case, the system is near the critical condition for three-phase separation, and the two coacervate phases are inclined to fuse with each other. The content of the A polymer in the III phase increased noticeably, and the interface between II and III phases becomes less distinguishable. Then, the interfacial width becomes much larger.

The interfacial tensions of the two interfaces have the opposite variation trend. As shown in Figure 4b, the change of interfacial tension with respect to σ_B follows an interesting

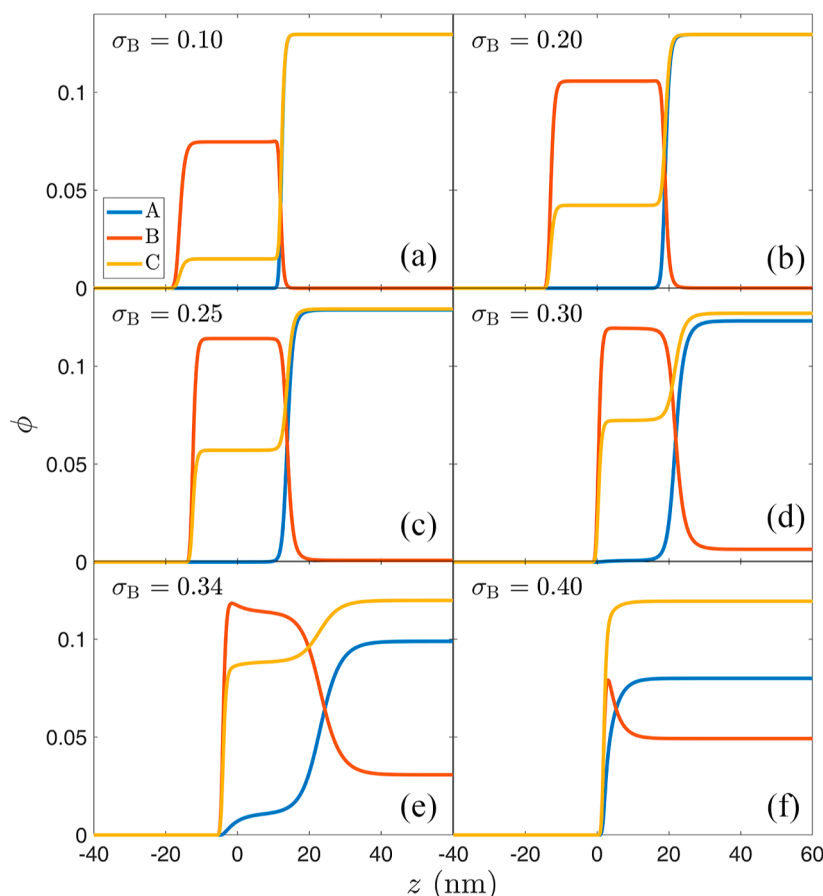


Figure 5. Density profiles of different species for systems with increasing σ_B . (a–e) Density profiles from the supernatant phase (I-phase) to the AC-rich coacervate phase (II-phase) after three-phase separation. (f) Density profile from the supernatant phase to the coacervate phase when the polyelectrolytes solution only separates into two phases.

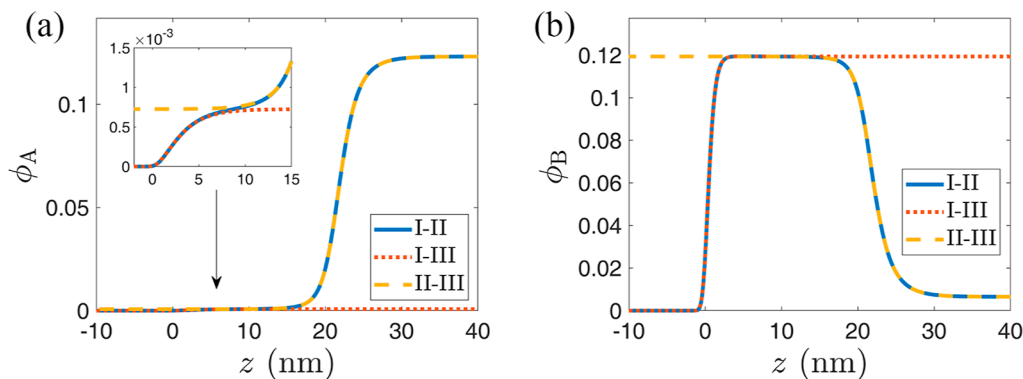


Figure 6. Density profiles from the I-phase to II-phase at $\sigma_B = 0.30$ (solid blue line), compared with those from the I-phase to III-phase (magenta dotted line) and from II-phase to III-phase (yellow dashed line).

linear relationship for both two interfaces. As σ_B increases, electrostatic interaction between B and C increases, inducing a stronger tendency for B molecules to participate in complexation with oppositely charged polymers. As a result, the interfacial tension between I and III phases increases. On the other hand, increasing σ_B causes less effective repulsion between polycations A and B and further leads to a decrease in the immiscibility between II and III phases as well as $\gamma^{\text{II-III}}$.

It is worth noting that the interfacial width (W/R_g) is approximately the same order of magnitude as the radius of gyration of polymers, indicating that the interface formed by complex coalescence of multicomponent polyelectrolytes is

relatively wide. eq 5 given in the theoretical section is based on the premise of a wide enough interface between two coexisted phases. Therefore, the results drawn from the calculation are in accord with the premise assumptions of the theory.

The two interfaces mentioned above show prospective and reasonable properties. However, beyond our expectation, the density profiles become rather complex when we explore the interfacial properties between the supernatant phase (I-phase) and the AC-rich coacervate phase (II-phase), as summarized in Figure 5. Here, one has to solve the interfacial problem with given boundary conditions on both sides, namely, the bulk concentrations of the I-phase are fixed in the leftmost region

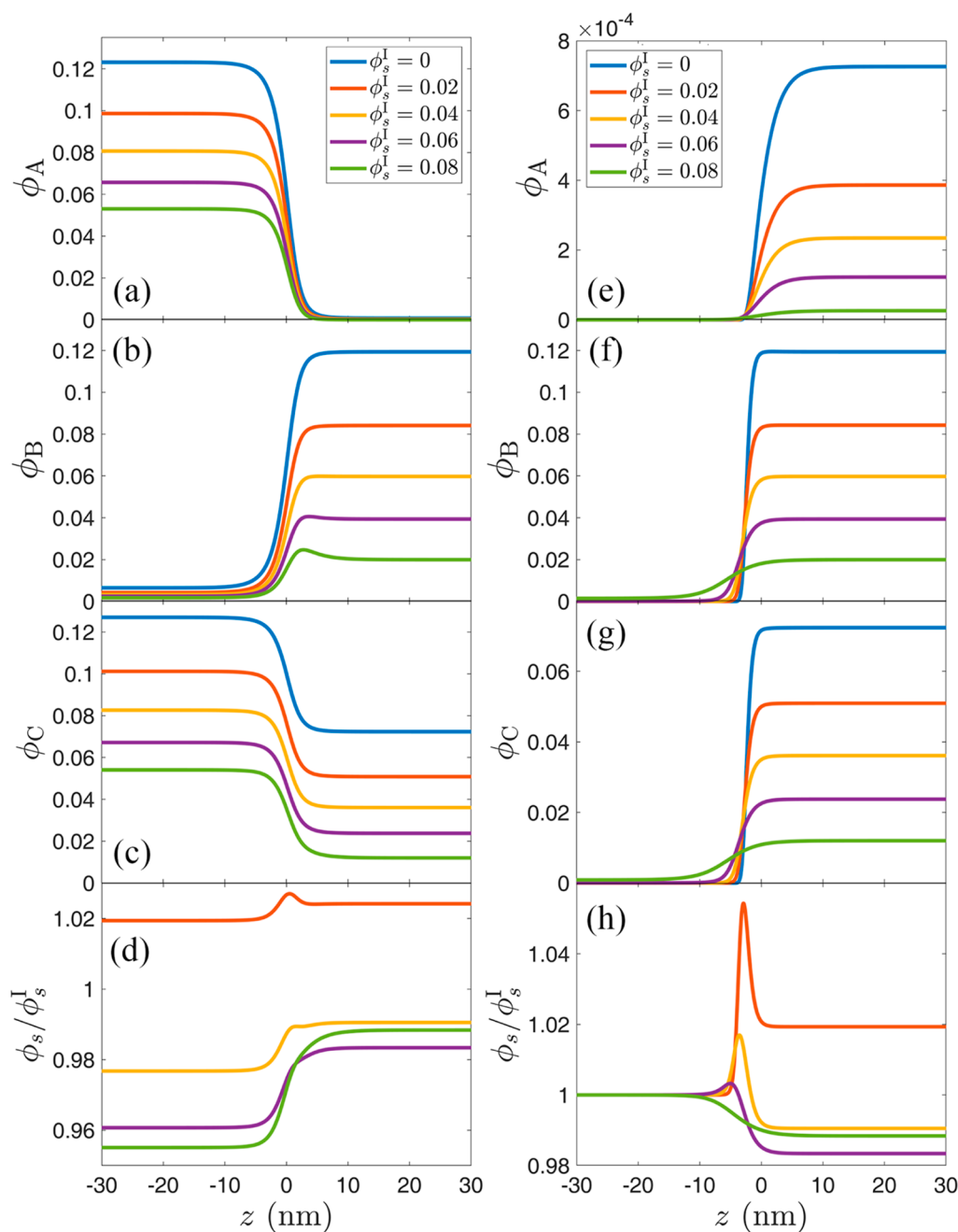


Figure 7. Density profiles of different species at different salt concentrations from the II-phase to III-phase (a–d) and from the I-phase to III-phase (e–h) for a system with fixed parameters of $\sigma_A = 0.50$, $\sigma_B = 0.30$, and $\sigma_C = -0.50$.

and the bulk concentrations of the II-phase are fixed in the rightmost region. The solution shows that the density profiles between phases I and phase II can be approximately divided into three regions, and a new astonished region emerges in the middle of these two bulk phases. From the supernatant phase to the AC-rich coacervate phase, the concentration of polycation A follows a monotonically increasing trend. But polycation B exhibits an unexpected non-monotonous behavior with a raised concentration plateau in the middle region, and the platform concentration is exactly equal to the bulk concentration of B monomers in phase III. The same plateau appears for polycation C with the concentration equal to the bulk concentration of C in phase III. We further compare the density profiles at the I–II interface with those at

the I–III interface and II–III interface, as depicted in Figure 6. Taking $\sigma_B = 0.30$ as an example, we find that the left branch of the density profiles across the I–II interface overlaps with that at the interface between I and III phases, while the right branch is in accord with the concentration curve at the interface of II–III phases completely. In addition, the interfacial tension between phases I and II, determined by the density profiles plotted in Figure 5, is approximately equal to the sum of γ^{I-II} and γ^{II-III} . This three-region density profile is independent of the initial guess of densities and represents a stable solution to the interface.

As mentioned above, the third phase emerging in the middle region actually represents phase III, which indicates that the region between the supernatant phase and the AC-rich

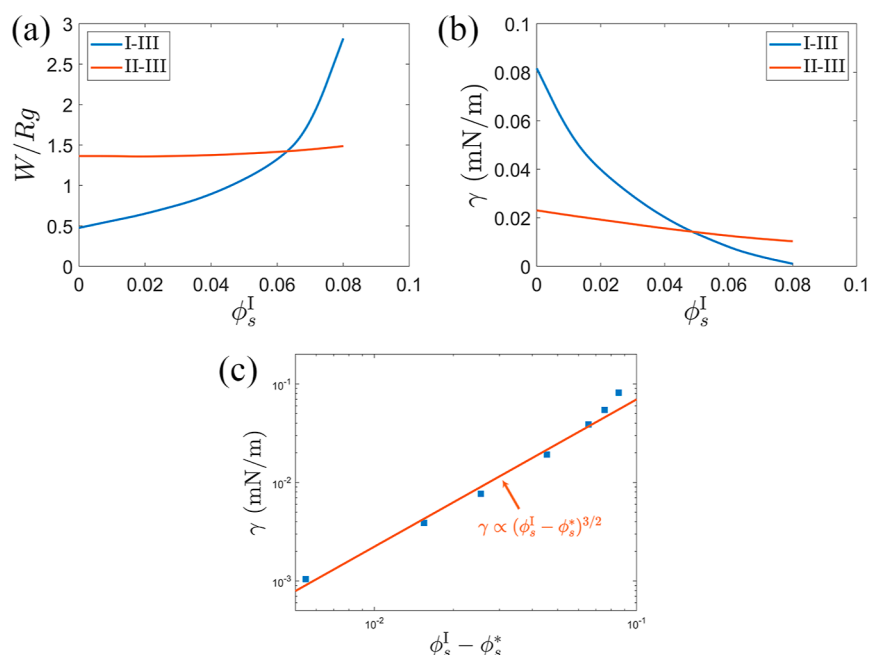


Figure 8. (a) Interfacial width between I and III phases (blue line) and II and III phases (magenta line) as a function of ϕ_s for the same system in Figure 7. Density profile of polyanion C is chosen to calculate the interfacial width and the vertical axis is plotted as the ratio of the interfacial width to the radius of gyration of the polyelectrolyte. (b) Interfacial tension as a function of ϕ_s . (c) Interfacial tensions between I and III phases γ^{I-III} as a function of the difference between supernatant salt concentration and the critical salt concentration $\phi_s^I - \phi_s^*$. Blue square symbols correspond to results obtained from theoretical prediction, with the red fit line showing the 1.5 scaling exponent near the critical point.

coacervate phase will exhibit a I–III–II multilayer structure. Therefore, we propose that a core–shell structure, where the BC-rich coacervate phase is sandwiched between the core of the AC-rich coacervate phase and the outer supernatant phase rather than a direct contact of the AC-rich coacervate phase and dilute phase, is favored when three-phase separation occurs in multicomponent systems.

We speculate that the underlying origin of the emergence of middle phase III between I and II phases is similar to the mechanism of enrichment of salt ions at the interface in the two-phase coacervation system, as well as the enrichment of weakly charged polycations B at the I/III phase interface in the multiphase coacervation system. Phase II is rich in polycation A of higher charge density. Direct contact of the II-phase and dilute I-phase will cause a considerably sharp interface because of the strong enough electrostatic correlation, which further induces significant chain conformation entropy loss and results in an extra free energy penalty. In this case, weaker charge polycation B in the system prefers to stay at the I/II phase interface to alleviate this entropy penalty. Continuous influx of B chains (also C chains due to electrostatic neutrality) into the interface finally leads to the formation of phase III in the middle region and finally leads to a three-layer nested structure of droplets-within-droplets. To verify this point, we consider a system with $\sigma_A = 0.50$, $\sigma_B = 0.40$, and $\sigma_C = -0.50$, in which the linear charge density asymmetry of A and B is inadequate to induce three-phase separation but the system separates into a coacervate phase of ABC mixture and a dilute phase. The accumulation of B chains at the interface can be observed obviously, as shown by a density peak at the interface in Figure S5f.

Therefore, if the multicomponent polyelectrolyte solution may undergo three-phase coacervation driven by the asymmetry of linear charge density, a core–shell structure

with a denser droplet constituting of higher charged polycations being wrapped by a droplet composed of lower charged polycations is preferentially formed. The three-layer nested structure formed in the multiphase separation process has been observed by some experiments,^{37,61} which represent quite probably the factual examples as our theoretical prediction.

Effects of Salt Concentration on Interfacial Properties. In this section, we further investigate the interfacial properties between different phases upon the addition of salt ions. The linear charge densities of the system are fixed as $\sigma_A = 0.50$, $\sigma_B = 0.30$, and $\sigma_C = -0.50$. When salt ions are added, the electrostatic screening effect complexes the distribution of all of the species. As the direct I–II interface is not stable, as mentioned in the last section, we will first consider the II–III interface and the I–III interface. The density profiles for each component across the two interfaces with varying salt concentrations are plotted in Figure 7.

At the interface between phases II and III, each species follows a slow distribution instead of any sharp variation. The concentrations of polyelectrolytes A and C showed a monotonic trend. The concentration of B changes monotonically at small salt concentrations but displays a global maximum value at the interface near the side of the III-phase at high salt concentration. On the contrary, the density profile of salt ions exhibits a peak at the interface at small salt concentration, and it gradually changes to a monotonic curve as the salt concentration increases. The enrichment of salt ions or less charged polyelectrolyte B at the interface has been discussed before. It is believed that the enrichment of charged species enhances the local electrostatic screening effect, lowering the extra free energy associated with the electrostatic correlation at the interface. It is interesting that the species enriched at the interface changes from salt ions to the less

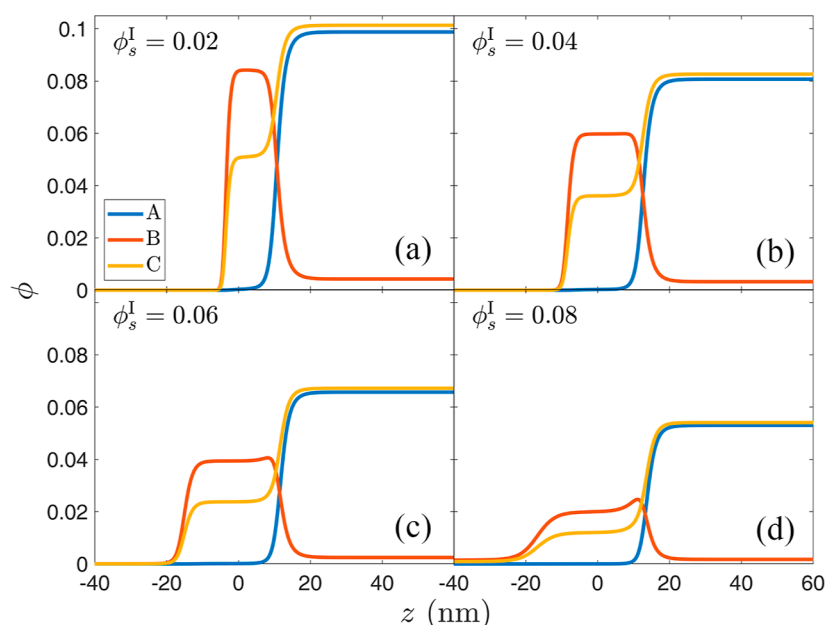


Figure 9. Density profiles of different species for systems with increasing ϕ_s from the supernatant phase (I-phase) to the AC-rich coacervate phase (II-phase) for the same system in Figure 7.

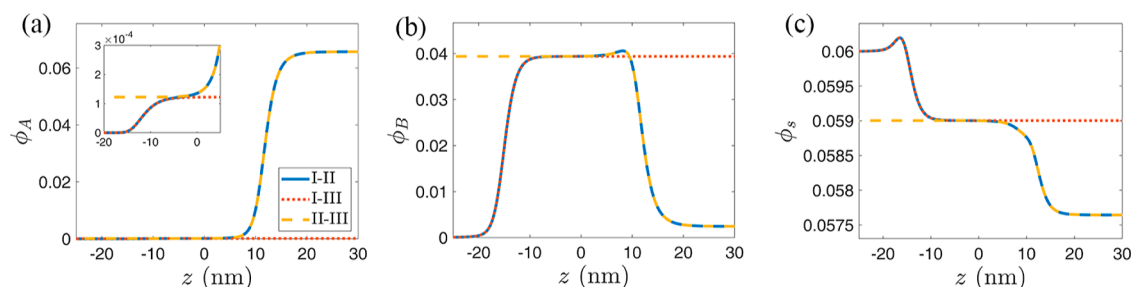


Figure 10. Density profiles from the I-phase to II-phase for the same system in Figure 7 at $\phi_s = 0.06$ (solid blue line), compared with those from the I-phase to III-phase (magenta dotted line) and from the II-phase to III-phase (yellow dashed line).

charged polycations B with increasing salt concentrations, as illustrated in Figure 7b,d. This change should be related to their relative concentration ratio at the interface peak to the bulk phase. The enrichment phenomenon is less impossible to observe as the concentration of some species is large enough. Another characteristic is that the partitioning of salt ions on the two bulk dense phases satisfies the same behavior as that of the simple two phase coacervation system; for example, the salt-partitioning coefficient is larger than 1 at low salt concentrations and is smaller than 1 at higher salt concentrations.

The density profiles from the I-phase to III-phase, depicted in Figure 7, are also similar to the previous results in the two-phase coacervation system. Generally, the density profiles of polyelectrolytes B and C are sharper than those of polyelectrolytes A although the content of the A monomer is rather low. All curves are flattened with the introducing of extra salt ions, and this is also a proxy of a broader interface. The peak value of salt concentration is observed, and then, it vanishes into a monotonic curve at higher salt concentrations (see the green line in Figure 7h).

Based on the density profiles, we further calculate the interfacial width and interfacial tension between phase III and phase I (or phase II) with varying salt concentrations. The results are listed in Figure 8. For the I–III interface, addition of

salt ions produces a stronger screening effect on the charges from oppositely charged polymers, which results in a weaker separation and a wider interface. As shown by the blue line in Figure 8, the interfacial width increases quickly and the interfacial tension decreases to zero, when the salt concentration approaches some critical point ϕ_s^* and the I–III interface gradually fades away. It is worth noting that the interfacial tension between I and III phases follows a power law dependence of $3/2$ on $\phi_s^I - \phi_s^*$ when approaching the critical point, which is equal to the scaling law of the two-phase coacervation system.

Nevertheless, the salt ions yield a rather weak influence on the properties of the II–III phase interface (see the magenta lines in Figure 8), although salt ions affect considerably the bulk II and III phases. Upon the addition of salt ions, the interfacial width remains nearly constant, and the decrease of interfacial tension follows a very slow rate compared with that of the I–III interface. This phenomenon is related to the effects of salt ions on the bulk phase behaviors. As our previous paper points out that salt ions appear to have little impact on the charge-asymmetry-driven immiscibility between A/C and B/C coacervates,⁵⁴ adding salt ions leads to the fusion of phase I and phase III but does not affect the separation of phase II and phase III. As Figure 7c shows, the distributions of polyanion C along the II/III interface display very similar

profiles accompanied by global reduction of polymer concentration with increasing salt concentration, which indicates a relatively stable difference between phase II and phase III. Note that at low salt concentration, the interfacial tension between II and III phases is even significantly smaller than that of the I/III interface, indicating the facile contact between two coacervates in the real system.

When we consider the density profiles across the I–II interface, it is demonstrated from Figure 9 that the density profile exhibits a pattern very similar to that in the absence of salt. The interface between I and II phases must evoke an intermediate phase, and it can be divided into three regions with the I-phase on the left, the II-phase on the right, and a new phase with a concentration plateau emerging in the middle region. Likewise, the concentration of each species in the middle region is almost equal to the corresponding bulk concentration value of phase III. The density profiles at the I–II interface are essentially equivalent to the combination of the profiles of the I–III interface and the III–II interface (see Figure 10), demonstrating the formation of a I–III–II double layer interface in the interfacial region of the I-phase and II-phase. It seems that the final structure of droplets is hardly affected by the salt ions, and we believe that it still tends to form the core–shell structure in multiphase coacervation even in the presence of salts, as shown in Figure 11.

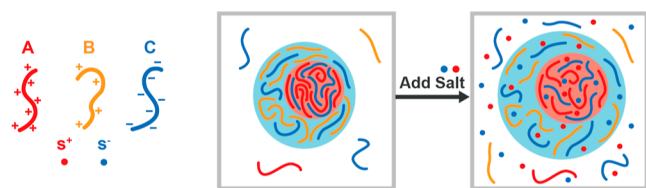


Figure 11. Core–shell structure formed in multiphase coacervation. The red region denotes the coacervate phase with higher charged density located in the inner part and the blue region denotes the shell comprised by weaker charge polymers. Addition of salt leads to some dissolution of coacervate phases but hardly affects the immiscibility between two coacervate phases and the core–shell structure.

SUMMARY

In this research, we investigated the interfacial properties in multiphase coacervation and systematically discussed the structure and architecture of multiphase droplets. For a three-phase coacervation system driven by charge density asymmetry, as shown in our previous paper,⁵⁴ the density profiles and interfacial tensions are calculated under the theoretical framework of RPA-based bulk free energy combined with the extra contribution arising from the density gradient.

For the three interfaces among the coexisting dilute phase and two condensed phases, our results showed that there is a linear relationship between interfacial tension and charge density asymmetry. Decreasing the charge density asymmetry between the two polycations results in a decrease in the interfacial tension $\gamma^{\text{II–III}}$ between the two coacervate phases but leads to an increase in $\gamma^{\text{I–III}}$ between the coacervate phase with weaker charge polymers and the dilute phase. An increase in salt concentration further leads to a 3/2 exponential decrease in $\gamma^{\text{I–III}}$. Addition of salt ions, however, has little effect on the interfacial properties between two coacervate phases, confirming that the immiscibility caused by charge density asymmetry

is hardly affected. Furthermore, our results suggest that a direct contact between the coacervate with higher charge density (phase II) and the dilute phase (phase I) is unstable, both in the presence and absence of salts. A middle layer consisting of a less condensed phase (phase III) must appear between them. Therefore, we speculated that a core–shell structure formed by the two coexisting coacervate phases is favored, where strong polycations with higher linear charge density tend to be located in the core part, whereas weakly charged polycations are enriched in the shell. Furthermore, the core–shell structure is stable regardless of the addition of salt ions as long as three-phase coacervation is maintained.

Despite some consistency between our predictions and experiments, we limit ourselves to elucidating the core–shell structure formed in multiphase coacervation. Other morphologies of small droplets, such as “Janus-particle” like structures, which have also been observed in experiments,⁶² are not described by this work. Other factors, such as the stoichiometry of the droplet phases and more driving forces including the hydrophobic interaction and excluded volume effect, are expected to be introduced in theory in order to fully understand the structure formed in multiphase coacervation. We adopt the weak inhomogeneity limit for simplicity, which may limit the accuracy for strongly segregated systems. Net charge inhomogeneity and the electric double layer effect are not considered at the interface in this work. Although the contribution from this effect may not be significant, we expect that other advanced theories incorporating the contribution of chain conformation and finite Galvani potential at the interface can give a more accurate description of interfacial properties in multiphase coacervation. Furthermore, other related factors, such as temperature, chain length, conformation, or other variations, remain to be clarified in further work.

Overall, our theoretical study extends the understanding of the interfacial properties of polyelectrolyte complexes from simple two-phase coacervation to multiphase coacervation. This may provide a physical mechanism to explain the multidroplet organization observed in experiments and offer further insights into understanding the LLPS and compartmentalization in biological systems.

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The authors are grateful for the financial support provided by the National Natural Science Foundation of China (NSFC) (Grant Nos. 22073002, 51921002, and 22373008).

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